Closure of the Dirichlet Tensor on Containers

Extracted from a machine-checked development

Abstract

We prove that the Dirichlet tensor $\otimes$ on the category **Cont** of containers is *closed*: for every container q the functor $- \otimes q$ has a right adjoint $\mathrm{Ihom}(q, -)$. The internal hom is itself a container, whose shapes are the container morphisms $q \Rightarrow r$ and whose positions over a morphism (u, g) are pairs consisting of a q-shape and an r-position over its image. The adjunction is exhibited by an explicit currying bijection of hom-sets, with both round trips and both naturalities holding by pure rearrangement of dependent products. The proofs of all stated results have been machine-checked; we record only the mathematical content.

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1 Introduction

Containers form a category **Cont** whose morphisms are pairs of a forward map on shapes and a backward map on positions. Among the monoidal structures on **Cont** is the *Dirichlet tensor*: for containers $C = (S, P)$ and $D = (T, Q)$,

$$C \otimes D \;=\; \big(S \times T,\ (s,t) \mapsto P(s) \times Q(t)\big). \tag{1}$$

Shapes are paired; positions are products. This is dual in spirit to the categorical product of containers, which uses the same shape product but coproducts of positions.

A monoidal category is closed when each functor $- \otimes q$ admits a right adjoint. In the polynomial-functor presentation this is classical (Niu–Spivak); the present note records the corresponding statement in the container presentation, with fully explicit currying and uncurrying maps and with naturality in both variables.

The material is elementary and self-contained. We work throughout in a single universe of types, writing **Type** for that universe. Morphisms of containers and the Dirichlet tensor are assumed as

in the prior notes on containers and monoidal structure; we recall only what is needed. Proofs are indicated by the phrase "by lean," meaning they have been checked by a proof assistant rather than written out in prose.

What this note claims. In ordinary mathematical language, the claim is: the Dirichlet monoidal structure $(\mathbf{Cont}, \otimes, y)$ is closed. The internal hom $\mathrm{Ihom}(q, r)$ is the container whose shape type is literally the set of container morphisms $q \Rightarrow r$, and the currying bijection

$$\mathrm{Hom}(p \otimes q,\, r) \;\cong\; \mathrm{Hom}\big(p,\, \mathrm{Ihom}(q, r)\big) \tag{2}$$

is pure currying of dependent products—split across the two components of the backward position map of a morphism $p \otimes q \Rightarrow r$. There is no extra data, no transport, and no exotic naturality constraint beyond what the container presentation already forces.

2 Recollections

Definition 2.1 (Container and morphism)**.** A **container** C consists of a type $\mathrm{Shape}(C)$ of shapes and, for each shape s, a type $\mathrm{Pos}(C)(s)$ of positions. A **morphism** $\varphi : C \Rightarrow D$ consists of a shape map $u : \mathrm{Shape}(C) \to \mathrm{Shape}(D)$ and, for each s, a position map $g_s : \mathrm{Pos}(D)(u(s)) \to \mathrm{Pos}(C)(s)$ (contravariant on positions). We write $\mathrm{Hom}(C, D)$ for the type of such morphisms.

Definition 2.2 (Dirichlet tensor)**.** Let $C = (S, P)$ and $D = (T, Q)$ be containers. Their **Dirichlet tensor** is the container

$$C \otimes D \;=\; \big(S \times T,\; (s, t) \mapsto P(s) \times Q(t)\big). \tag{3}$$

The unit y is the one-shape, one-position container (the same object that serves as unit for sequential composition of containers).

Definition 2.3 (Bifunctorial action)**.** Given morphisms $\mu : C \Rightarrow C'$ and $\nu : D \Rightarrow D'$, the **tensor of morphisms** $\mu \otimes \nu : C \otimes D \Rightarrow C' \otimes D'$ acts componentwise:

$$(\mu \otimes \nu) \text{ on shapes:} \quad (s, t) \mapsto \big(u_\mu(s),\, u_\nu(t)\big), \tag{4}$$

$$(\mu \otimes \nu) \text{ on positions:} \quad (p, q) \mapsto \big(g_\mu(p),\, g_\nu(q)\big). \tag{5}$$

3 The internal hom

The internal hom for $\otimes$ is a container whose shapes *are* morphisms. That is not a notational convenience: it is the content of the closed structure (cf. Niu–Spivak, Eq. (4.79), transported to containers).

Definition 3.1 (Internal hom)**.** Let q and r be containers. The **internal hom** $\mathrm{Ihom}(q, r)$ is the container defined by

$$\mathrm{Shape}\big(\mathrm{Ihom}(q, r)\big) \;=\; \mathrm{Hom}(q, r), \tag{6}$$

$$\mathrm{Pos}\big(\mathrm{Ihom}(q, r)\big)(f) \;=\; \sum_{t \,:\, \mathrm{Shape}(q)} \mathrm{Pos}(r)\big(u_f(t)\big), \tag{7}$$

where $f = (u_f, g_f) : q \Rightarrow r$. Thus a shape of $\mathrm{Ihom}(q, r)$ is a container morphism $q \Rightarrow r$, and a position over such a shape is a pair $\langle t,\, i \rangle$ consisting of a q-shape t and an r-position i over the image shape $u_f(t)$.

Proposition 3.2 (Shapes are morphisms)**.** *For all containers q and r,*

$$\mathrm{Shape}\big(\mathrm{Ihom}(q,r)\big) \;=\; \mathrm{Hom}(q,r) \tag{8}$$

definitionally.

Proof. By lean. □

Proposition 3.3 (Positions of the internal hom)**.** *For every morphism $f : q \Rightarrow r$,*

$$\mathrm{Pos}\big(\mathrm{Ihom}(q,r)\big)(f) \;=\; \sum_{t\,:\,\mathrm{Shape}(q)} \mathrm{Pos}(r)\big(u_f(t)\big) \tag{9}$$

definitionally.

Proof. By lean. □

Remark 3.4. That the shape type is the hom-set of **Cont** is what makes the adjunction *internal*: the internal hom is a container, not merely a set of morphisms. The extra data—the position family—is precisely what is needed for the currying bijection of the next section to be a bijection of container morphisms rather than of bare functions.

4 Currying and uncurrying

Unfolded, a morphism $F : p \otimes q \Rightarrow r$ consists of

- a shape map $u_F : \mathrm{Shape}(p) \times \mathrm{Shape}(q) \to \mathrm{Shape}(r)$, and
- a position map $g_F : (s,t) \to \mathrm{Pos}(r)\big(u_F(s,t)\big) \to \mathrm{Pos}(p)(s) \times \mathrm{Pos}(q)(t)$.

A morphism $G : p \Rightarrow \mathrm{Ihom}(q,r)$ consists of

- a shape map $u_G : \mathrm{Shape}(p) \to \mathrm{Hom}(q,r)$, sending each s to a pair (u_s, φ_s), and
- a position map $g_G : (s : \mathrm{Shape}(p)) \to \big(\sum_t \mathrm{Pos}(r)(u_s(t))\big) \to \mathrm{Pos}(p)(s)$.

The bijection is pure currying, *split across the two components of g_F*: the $\mathrm{Pos}(q)(t)$ component of g_F becomes the backward position map φ_s of the shape $u_G(s)$, and the $\mathrm{Pos}(p)(s)$ component becomes g_G. That split is the whole content—it is why the positions of $\mathrm{Ihom}(q,r)$ are $\sum_t \mathrm{Pos}(r)(u(t))$ and not something else.

Definition 4.1 (Currying)**.** Let $F : p \otimes q \Rightarrow r$ be a container morphism. The **curried morphism**

$$\mathrm{curry}(F) \;:\; p \;\Rightarrow\; \mathrm{Ihom}(q,r) \tag{10}$$

is defined by

$$u_{\mathrm{curry}(F)}(s) \;=\; \big(t \mapsto u_F(s,t),\; (t,i) \mapsto \pi_2\big(g_F(s,t)(i)\big)\big), \tag{11}$$

$$g_{\mathrm{curry}(F)}\big(s,\, \langle t,i\rangle\big) \;=\; \pi_1\big(g_F(s,t)(i)\big). \tag{12}$$

Thus the q-component of F's backward position map is packaged *into the shape* of the internal hom, while the p-component becomes the backward position map of the curried morphism.

Definition 4.2 (Uncurrying)**.** Let $G : p \Rightarrow \mathrm{Ihom}(q, r)$ be a container morphism. The **uncurried morphism**

$$\mathrm{uncurry}(G) \;:\; p \otimes q \;\Rightarrow\; r \tag{13}$$

is defined by

$$u_{\mathrm{uncurry}(G)}(s, t) \;=\; u_{u_G(s)}(t), \tag{14}$$

$$g_{\mathrm{uncurry}(G)}\big((s, t),\, i\big) \;=\; \Big(g_G\big(s,\, \langle t, i\rangle\big),\; g_{u_G(s)}(t, i)\Big). \tag{15}$$

Read the shape map of G at s as a morphism $q \Rightarrow r$; its shape map supplies the r-shape, and its backward position map supplies the $\mathrm{Pos}(q)(t)$ component. The $\mathrm{Pos}(p)(s)$ component is G's own backward position map applied to the position $\langle t, i\rangle$.

Theorem 4.3 (Uncurry after curry)**.** *For every morphism $F : p \otimes q \Rightarrow r$,*

$$\mathrm{uncurry}\big(\mathrm{curry}(F)\big) \;=\; F. \tag{16}$$

Proof. By lean. □

Theorem 4.4 (Curry after uncurry)**.** *For every morphism $G : p \Rightarrow \mathrm{Ihom}(q, r)$,*

$$\mathrm{curry}\big(\mathrm{uncurry}(G)\big) \;=\; G. \tag{17}$$

Proof. By lean. □

Remark 4.5 (Why both round trips are immediate)**.** Both identities are rearrangements of a $\Sigma/\times$ telescope. The only ingredients are η for products (rebuilding a pair from its projections), η for dependent sums (rebuilding $\langle t, i\rangle$), and η for the structure of a container morphism (rebuilding a morphism from its shape and position components). No transport of paths, no function extensionality beyond definitional η, and no appeal to naturality are required. The same definitional character holds for the monoidal coherences of $\otimes$ itself: the Dirichlet tensor is well behaved at the level of definitional equality in a way that sequential composition and the directed container comonad laws are not.

5 The closure theorem

Definition 5.1 (Type equivalence)**.** A **type equivalence** $A \simeq B$ is a pair of maps $f : A \to B$ and $g : B \to A$ together with proofs that $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$.

Theorem 5.2 (Dirichlet closure)**.** *For all containers p, q, and r there is a type equivalence*

$$\mathrm{Hom}(p \otimes q,\, r) \;\simeq\; \mathrm{Hom}\big(p,\, \mathrm{Ihom}(q, r)\big), \tag{18}$$

with forward map curry *and inverse* uncurry*. The two round-trip identities are Theorems 4.3 and 4.4. Equivalently,*

$$- \otimes q \;\dashv\; \mathrm{Ihom}(q, -). \tag{19}$$

Proof. By lean. □

Remark 5.3. Together with the monoidal structure $(\mathbf{Cont}, \otimes, y)$—associator, unitors, pentagon, and triangle—this makes $(\mathbf{Cont}, \otimes, y)$ a **closed monoidal category**, with internal hom Ihom.

6 Functoriality of the internal hom and naturality

The currying bijection is natural in p (contravariantly, by precomposition) and in r (covariantly, by postcomposition). Naturality in r requires the action of $\mathrm{Ihom}(q, -)$ on morphisms, which we record first.

Definition 6.1 (Action on morphisms)**.** Let $\psi : r \Rightarrow r'$ be a container morphism. The induced morphism

$$\mathrm{Ihom}(q, \psi) \;:\; \mathrm{Ihom}(q, r) \;\Rightarrow\; \mathrm{Ihom}(q, r') \tag{20}$$

is defined by

$$\text{on shapes:} \quad f \mapsto f \circ \psi \qquad \text{(postcomposition)}, \tag{21}$$

$$\text{on positions:} \quad \langle t,\, i\rangle \mapsto \langle t,\; g_\psi(u_f(t))(i)\rangle. \tag{22}$$

A position $\langle t, i\rangle$ of $\mathrm{Ihom}(q, r')$ over the shape $f \circ \psi$ has i an r'-position over ψ's image of $u_f(t)$; pulling i back through g_ψ yields an r-position, hence a position of $\mathrm{Ihom}(q, r)$.

Proposition 6.2 (Functoriality of $\mathrm{Ihom}(q, -)$)**.** *For every container q and every container r,*

$$\mathrm{Ihom}(q, \mathrm{id}_r) \;=\; \mathrm{id}_{\mathrm{Ihom}(q,r)}. \tag{23}$$

For morphisms $\psi : r \Rightarrow r'$ and $\psi' : r' \Rightarrow r''$,

$$\mathrm{Ihom}(q,\, \psi \circ \psi') \;=\; \mathrm{Ihom}(q, \psi) \circ \mathrm{Ihom}(q, \psi'). \tag{24}$$

Proof. By lean. □

Theorem 6.3 (Naturality in p)**.** *Let $h : p' \Rightarrow p$ and $F : p \otimes q \Rightarrow r$. Then*

$$\mathrm{curry}\big((h \otimes \mathrm{id}_q) \circ F\big) \;=\; h \circ \mathrm{curry}(F). \tag{25}$$

Proof. By lean. □

Theorem 6.4 (Naturality in r)**.** *Let $F : p \otimes q \Rightarrow r$ and $\psi : r \Rightarrow r'$. Then*

$$\mathrm{curry}(F \circ \psi) \;=\; \mathrm{curry}(F) \circ \mathrm{Ihom}(q, \psi). \tag{26}$$

Proof. By lean. □

Proposition 6.5 (Naturality of uncurry in p)**.** *Let $h : p' \Rightarrow p$ and $G : p \Rightarrow \mathrm{Ihom}(q, r)$. Then*

$$\mathrm{uncurry}(h \circ G) \;=\; (h \otimes \mathrm{id}_q) \circ \mathrm{uncurry}(G). \tag{27}$$

Proof. By lean. □

7 The counit: evaluation

Every adjunction has a counit. For Dirichlet closure the counit is evaluation: applying a morphism stored as a shape of the internal hom.

Definition 7.1 (Evaluation)**.** The **evaluation morphism**

$$\mathrm{ev}_{q,r} \;:\; \mathrm{Ihom}(q,r) \otimes q \;\Rightarrow\; r \tag{28}$$

is obtained by uncurrying the identity on $\mathrm{Ihom}(q,r)$:

$$\mathrm{ev}_{q,r} \;=\; \mathrm{uncurry}\big(\mathrm{id}_{\mathrm{Ihom}(q,r)}\big). \tag{29}$$

Unfolded, it sends the shape (f,t) to $u_f(t)$—that is, it *applies* the morphism stored in the shape—and its backward position map splits an r-position i into the $\mathrm{Ihom}(q,r)$-position $\langle t,i\rangle$ and the q-position $g_f(t,i)$.

Theorem 7.2 (Triangle identity / universal property)**.** *For every morphism $F : p \otimes q \Rightarrow r$,*

$$\big(\mathrm{curry}(F) \otimes \mathrm{id}_q\big) \circ \mathrm{ev}_{q,r} \;=\; F. \tag{30}$$

Proof. By lean. □

Theorem 7.3 (Uniqueness of the curried mate)**.** *Let $F : p \otimes q \Rightarrow r$ and $G : p \Rightarrow \mathrm{Ihom}(q,r)$. If*

$$(G \otimes \mathrm{id}_q) \circ \mathrm{ev}_{q,r} \;=\; F, \tag{31}$$

then $G = \mathrm{curry}(F)$.

Proof. By lean. □

Remark 7.4. Existence (Theorem 7.2) is the statement that $\mathrm{curry}(F)$ is a mate for F under evaluation. Uniqueness (Theorem 7.3) is the observation that any G satisfying the triangle is recovered as curry of its composite with evaluation, which is definitionally $\mathrm{uncurry}(G)$, so the claim reduces to the round trip of Theorem 4.4.

8 Discussion

What has been established. The Dirichlet tensor on containers is closed. The internal hom has shapes equal to container morphisms and positions equal to pairs of a source shape with a target position over its image. Currying is a pure split of the backward position map of a morphism out of a tensor. Both round trips, both naturalities, functoriality of $\mathrm{Ihom}(q,-)$, and the evaluation triangle hold by definitional rearrangement.

Relation to the polynomial presentation. In the polynomial-functor ("Poly") presentation of Niu–Spivak, the same adjunction is Eq. (4.79) of *Polynomial Functors*. The container presentation makes the split of position data completely explicit: one sees exactly which component of the tensor's position product becomes shape data of the internal hom and which becomes its position data.

Definitional character. Every proof in the development is by reflexivity of equality after unfolding definitions. This is a feature of the Dirichlet tensor and of container morphisms: both are built from products and dependent sums in a way that definitional η identifies the relevant telescopes. By contrast, sequential composition of containers and the directed-container comonad laws require genuine transports.

Scope. The development treats a single universe of types and does not address size issues or universe polymorphism. It assumes the monoidal structure of $\otimes$ (associator, unitors, coherences) from the companion note on the Dirichlet tensor; those coherences are likewise definitional.

Machine checking. Every definition and theorem statement above corresponds to a declaration in a machine-checked formalisation. The phrase "by lean" indicates that the proof has been verified by a proof assistant; we have not reproduced the formal proof scripts here. A mathematician reading this note need only verify that the English statements match the intended mathematics; the formal system is responsible for the derivations.

Acknowledgements

The mathematical content follows the theory of polynomial functors and their Dirichlet monoidal structure as developed by Niu and Spivak, transported to the container presentation of Abbott, Altenkirch, and Ghani. The presentation here is a prose rendering of a formal development intended for human review of the theorem statements.

References

[1] M. Abbott, T. Altenkirch, and N. Ghani. Containers: constructing strictly positive types. *Theoretical Computer Science*, 342(1):3–27, 2005.

[2] N. Niu and D. I. Spivak. Polynomial functors: a mathematical theory of interaction. arXiv:2312.00990, 2023.